

ZEERBOX · DESIGN DECISIONS REGISTER

ZeerBox design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for open items 1 to 10 on 2026-10-02 (ZBX-DDR-003 accepted with A1 and A2, fired brick, MIT D-Lab and Mali as first candidates, appearance items); moved to decisions made
0.1	2026-10-02	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

ZeerBox design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The fan frame is no wider than 290 mm and fits a 250 mm hole	The fan box bay is 310 mm between joists	ZBX-DDR-003, C1
2	The fan curve and the pad's pressure drop and effectiveness from supplier data	The airflow of about 560 m ³ /h and the 10.7 K drop rest on assumed curves	ZBX-CAL-001, sections 1 and 3
3	The panel frame has mounting holes on its back flange that the top rails can reach	The panel bolts to the mounting frames through them	ZBX-DDR-003, C8
4	Hurricane ties are rated 2 kN or more in uplift	Each purlin crossing sees about 1.1 kN at a 30 m/s gust	ZBX-CAL-001, Table 6a
5	A durable, pest-proof rice husk supply near the site; otherwise dry sand	The fill's insulation and the wall heat gain depend on it	ZBX-DDR-002, item 12
6	The local design wind speed, confirmed with a builder	The roof footings and ties are sized for a 30 m/s gust	ZBX-DDR-002, item 13
7	The pump gives about 3.6 L/min at 2 m head and has a float switch	The pad must stay wet and the pump must not run dry	ZBX-CAL-001, section 8
8	The charge controller is set for 12.8 V lithium iron phosphate and the BMS takes 10 A of charge	Battery safety and life	ZBX-CAL-001, section 7

■ Value engineering

Value-engineering target: USD 300 for the cooling equipment kit (a hypothetical control target, not a limit).

Estimated cost of the constructable design: USD 299 for the kit (USD 1 under the target). The structure is costed separately at USD 574, and the whole store at USD 873, crates excluded. Main cost drivers and savings worth trying:

- The largest kit lines are the 150 W panel (USD 62), the power box with its battery (USD 62), the pad with its frame, header and gutter (USD 50), the two fans (USD 30) and the controller (USD 30).
- Making the design constructable added USD 22 for the two panel mounting frames and USD 10 for the pad frame, lining, bars and brackets, less USD 3 for the roof clamps they replace: the kit rose from USD 270 to USD 299.

- The structure rose from USD 355 to USD 574 because the concept did not price the strip footing (USD 35), the roof beams and purlins (USD 70) or the door lining and fan box (USD 25), and underpriced the framed ceiling and the racks.
- Savings worth trying: galvanized steel angle in place of aluminium for the panel frames (about USD 8 less, needs painting at cut ends); a 100 W panel is not one (6 % short on energy, ZBX-CAL-001 section 7); in the structure, mud brick or stabilized block if decision 4 allows, and racks of local poles in place of sawn timber.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 8: the USD 300 budget covers the cooling kit with the structure costed separately; 150 W panel; keep the battery; forced air through a pad; exhaust fans with the pad opposite; wet sand cavity pending the heat balance; 24 crates; control thresholds as starting values	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	ZBX-DDR-001
2026-09-25	Items 11 to 13: 150 mm pad, R3 relaxed to 80 % with covered or lined crates; dry rice husk fill in place of wet sand (dry sand as fallback); 500 x 500 x 600 mm roof footings as a minimum	Amish: "i accept all your recommendations, go with them across all repos."	ZBX-DDR-002
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	ZBX-DDR-003 (the changes themselves await his review, open decision 1)
2026-09-30	Open decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-01	budget_usd is a value-engineering target, not a spending limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	ZBX-CAL-001 v0.3, section 11
2026-10-02	Design for construction accepted: the changes C1 to C10 of ZBX-DDR-003, as made	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	ZBX-DDR-003, Table 1
2026-10-02	Fan position: the two fans stay in the ceiling fan box at the door end, as modelled	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	ZBX-DDR-003, A1
2026-10-02	Exhaust into the roof space accepted, with a check at TRL 4 that moist air does not drift back to the pad; a short duct past the roof edge is added only if it does	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	ZBX-DDR-003, A2
2026-10-02	Wall material: fired brick, as modelled and calculated	Amish: "APPROVED: The open decisions from the last wave	ZBX-DDR-001, item 9

Date	Decision	Decided by	Record
		(TremorTrace to ZeerBox) came in after the review and aren't on the page either."	
2026-10-02	First co-design partner, region and crop mix: a group that has already field-tested evaporative cooling chambers with vegetable growers; the first candidate to approach is MIT D-Lab's evaporative cooling team, with a dry-season vegetable area of Mali as the first candidate region and tomatoes, peppers and okra as the crop mix, leafy greens on a separate rack	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	ZBX-DDR-001, item 10
2026-10-02	Clear window in the controller lid only; the power box has a plain door	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, item 2
2026-10-02	Mode lamp on top of the controller box beside the door, added to the model	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, item 3
2026-10-02	Outside sensor shield: stacked round plates 46 mm across	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, item 4
2026-10-02	Plinth render band: an optional 10 mm finish 300 mm high, the mason's choice	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, item 5
2026-10-02	Hose and conduit routing accepted as sketched in the build plan, routed on site	Amish: "APPROVED: The open decisions from the last wave (TremorTrace to ZeerBox) came in after the review and aren't on the page either."	REVIEW.md, 2026-09-26, item 6